

Fluid Flow Operation

Text Books:

- **K.A. Gavanhe, “Unit Operations - I”, Nirali Prakashan, 2015**
- W. L. McCabe, J.C. Smith and P. Harriott, “Unit operations of Chemical Engineering”, McGraw Hill, International Edn., 7th Edition, 2005.

Unit 5: Transportation and Metering

Measurement of fluid flow:

- Venturi meter
- Orifice meter
- Pitot tube
- Rotameter
- Weirs and notches
- Wet gas meter and dry gas meter
- Hot wire and hot film anemometers

Measurement of fluid flow

- In the chemical process industry, it is desirable to know the amount of a fluid flowing to or from the process equipment.
- Many different types of flow meters are used industrially to measure the rate at which a fluid is flowing through a pipe or a duct.
- The flow measuring devices or flow meters are classified as:
 1. Variable head meters, e.g., orifice meter, venturi meter.
 2. Variable area meters, e.g., rotameter.
 3. Current meters, e.g., cup type current meter.
 4. Positive displacement meters, e.g., wet gas meter.
 5. Electromagnetic meters, e.g., magnetic meter.

Measurement of fluid flow

- In the **variable head meters** (orifice meter or venturi meter), the area of constriction or area of flow is constant and the pressure drop across the meter varies with the flow rate or discharge
- While in the **variable area meter**, the pressure drop across the meter is constant and the flow rate is a function of the area of constriction or area of flow.

Venturi Meter

- **Principle :** The basic principle on which a venturi meter works is that by reducing the cross-sectional area of the flow passage, a pressure difference is created and the measurement of the pressure difference (between the inlet of the meter and at a point of reduced pressure) enables the estimation of the discharge/flow rate through the pipe.
- A venturi meter is a variable head meter which is used for measuring the flow rate of a fluid through a pipe.
- In this meter, the fluid is gradually accelerated to a throat and then gradually retarded in a diverging section where the flow expands to the pipe size.
- A large portion of the kinetic energy is thus recovered (converted back to pressure energy).

Venturi Meter

Construction

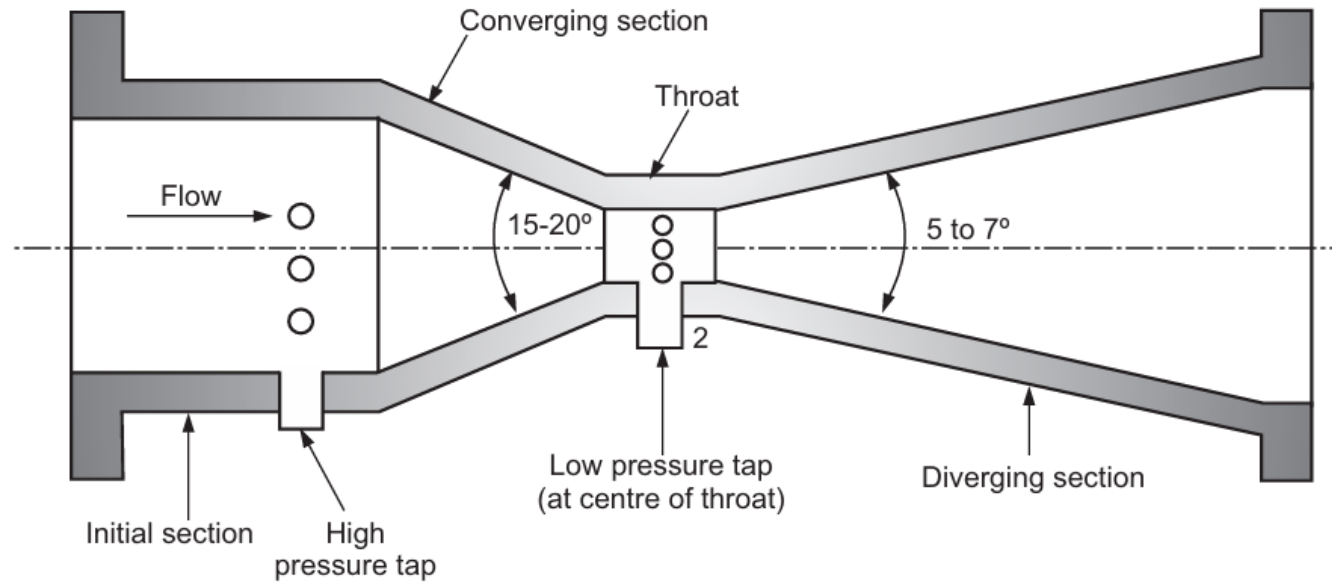
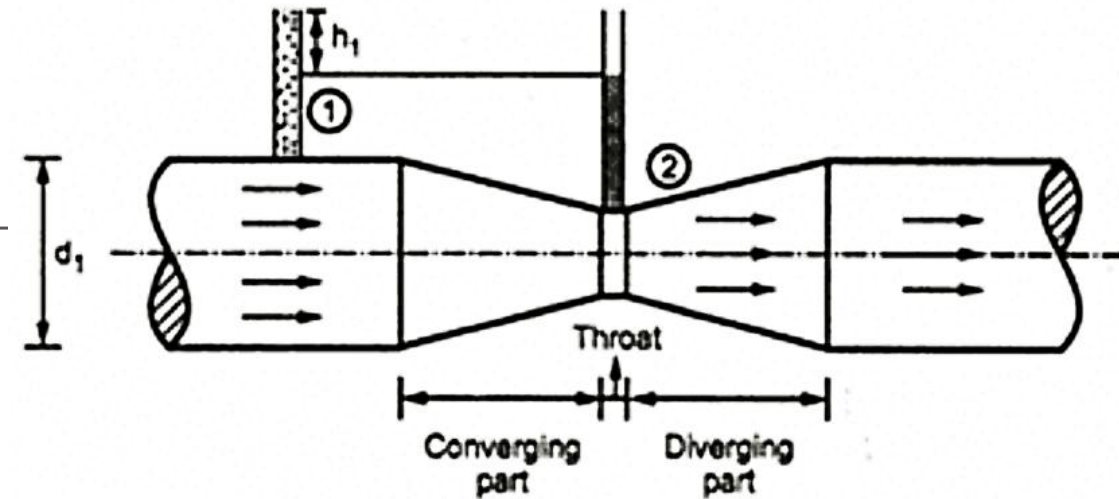


Fig. 7.22 : Venturi meter



Venturi Meter

- **A venturi meter consists of**

1. **An inlet section followed by a convergent cone/section:** The inlet section of the venturi meter is of the same diameter as that of the pipe line followed by the short convergent section with a converging cone angle of $15-20^\circ$ and its length parallel to the axis is approximately equal to $2.7 (D - D_T)$, where D is the pipe diameter and D_T is the throat diameter. In the convergent section, the fluid is accelerated.
2. **A cylindrical throat:** the section of constant cross-section with its length equal to diameter. The flow area is minimum at the throat. In general, the diameter of throat may vary from $1/3$ to $3/4$ of the pipe diameter and more commonly the diameter of throat is $1/2$ the pipe diameter.
3. **Gradual divergent cone:** with a cone angle of about $5 - 7^\circ$ wherein the fluid is retarded and a large portion of the kinetic energy is converted back into the pressure energy.

At the inlet section and the throat of the venturi meter, pressure taps are provided through the pressure rings/piezometer rings.

Venturi Meter

- In the meter, the fluid is accelerated in the converging cone and then retarded in the diverging cone gradually.
- An increase in the flow velocity at the throat (a section of minimum flow area) results in a decrease in the pressure at the throat.
- Due to this a pressure difference is developed between the inlet section and the throat section which is measured/noted with the help of manometer or pressure gauges (connected to the pressure taps) after the steady state is attained.
- This pressure difference/drop is then related to the flow rate by a mathematical flow equation for the meter.
- An equation for the drop in pressure due to venturi effect may be derived from a combination of Bernoulli's principle and the equation of continuity.

Venturi Meter

Let d_1 = Diameter at inlet or at section 1,

V_1 = velocity of fluid at section 1

$$a_1 = \text{Area at inlet} = \frac{\pi d_1^2}{4}$$

P_1 = Pressure at section 1

and d_2 , V_2 , a_2 and P_2 are the corresponding values at section 2.

Applying Bernoulli's equations at section 1 and section 2, we get,

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{v_2^2}{2g} + z_2 \text{ ----- (1.1)}$$

Since the pipe is horizontal, so $z_1 = z_2$

$$\frac{P_1}{\rho g} + \frac{v_1^2}{2g} = \frac{P_2}{\rho g} + \frac{v_2^2}{2g}$$

Venturi Meter

$$\frac{P_1 - P_2}{\rho g} = \frac{v_2^2 - v_1^2}{2g} \text{ ----- (1.2)}$$

But $\frac{P_1 - P_2}{\rho g}$ difference of pressure head at section 1 and 2 = h ----- (1.3)

substituting this value of $\frac{P_1 - P_2}{\rho g} = h$ in equation (1.2)

$$h = \frac{v_2^2 - v_1^2}{2g} \text{ ----- (1.4)}$$

Now applying continuity equation at section 1 and 2

$$a_1 v_1 = a_2 v_2 \quad \text{or} \quad v_1 = \frac{a_2 v_2}{a_1}$$

Substituting value of v_1 in equation (1.4) we get

$$h = \frac{v_2^2}{2g} - \frac{\left(\frac{a_2 v_2}{a_1} \right)^2}{2g}$$

Venturi Meter

$$= \frac{v_2^2}{2g} - \frac{(a_2^2 v_2^2)}{a_1^2 2g}$$

$$= \frac{v_2^2}{2g} \left(1 - \frac{a_2^2}{a_1^2}\right)$$

$$v_2^2 = 2gh \left(\frac{a_1^2}{a_1^2 - a_2^2}\right)$$

$$v_2 = \sqrt{2gh \left(\frac{a_1^2}{a_1^2 - a_2^2}\right)}$$

$$v_2 = 2gh \left(\frac{a_1^2}{a_1^2 - a_2^2}\right)$$

$$v_2 = \frac{a_1}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

$$\left(h = x \left(1 - \frac{\rho_L}{\rho}\right)\right)$$

Where,

x = difference between the liquid column in U tube,

ρ_L = density of lighter liquid,

ρ = density of liquid flowing through pipe.

Venturi Meter

But, discharge through venturimeter,

$$Q = a_2 v_2$$

$$Q_{th} = \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh} \text{ ----- (1.5)}$$

$Q = C\sqrt{h}$, where C = constant of venturi meter.

$$= \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2g}$$

Equation (1.5) gives the discharge under ideal conditions and is called as theoretical discharge.

Actual discharge is given by,

Actual discharge = Coefficient of venturimeter x Theoretical discharge

$$Q_{ac} = c_d \frac{a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gh}$$

Venturi Meter

$$Q_{ac} = C_d A_1 A_2 \sqrt{\frac{2gh}{A_1^2 - A_2^2}}$$
$$h = \frac{P_1 - P_2}{\rho g}$$
$$\beta = \frac{d_1}{d_2}$$
$$Q_{ac} = C_d A_2 \sqrt{\frac{2g \times \frac{P_1 - P_2}{\rho g}}{1 - \frac{A_2^2}{A_1^2}}}$$
$$Q_{ac} = C_d A_2 \frac{\sqrt{\frac{2(P_1 - P_2)}{\rho}}}{\sqrt{1 - \beta^4}}$$

Venturi Meter

- The coefficient of discharge (C_d) of a venturi meter is the ratio of the actual discharge to the theoretical discharge of the venturi meter.
- The value of theoretical discharge through the venturi as it has been obtained by considering no frictional losses in the system.
- For a well designed venturimeter, the coefficient of venturi, **C_d is about 0.98** for pipe diameters ranging from 50 to 200 mm and about 0.99 for larger sizes.
- Since the angle of divergence in the recovery cone (i.e., in the divergent cone) is small, the permanent pressure loss from a venturi meter is relatively small.
- The pressure recovery in a venturi meter is very high and thus may be used where only a small pressure head is available though it is expensive.

Venturi Meter

Advantages of Venturi meter :

1. Low permanent pressure loss and hence high pressure recovery.
2. High accuracy over wide flow ranges.
3. It can be used for flow measurement of compressible and incompressible fluids.
4. It can also be used where only a small pressure head is available.
5. High reproducibility.
6. Less power loss.

Disadvantages of Venturi meter :

1. It is expensive and bulky.
2. It occupies considerable space.
3. Relatively complex in construction.
4. The ratio of throat diameter to pipe diameter cannot be changed.
5. Not suitable for flow measurement of highly viscous slurries.
6. Used only for permanent installations.
7. It cannot be altered once it is installed.

Orifice Meter

Principle :

- In this meter, the fluid is accelerated by causing it to flow through a sudden constriction (orifice), the kinetic energy of the fluid increases and the pressure energy therefore decreases.
- With this meter, the overall pressure drop across the meter is high - a large percentage of the pressure drop across the meter is not recoverable, but is relatively cheap and reliable instrument and its installation requires a small length as compared to the venturi meter.
- Because of this where the space is limited, the orifice meter may be used for the measurement of discharge (flow rate) through pipes.

Orifice Meter

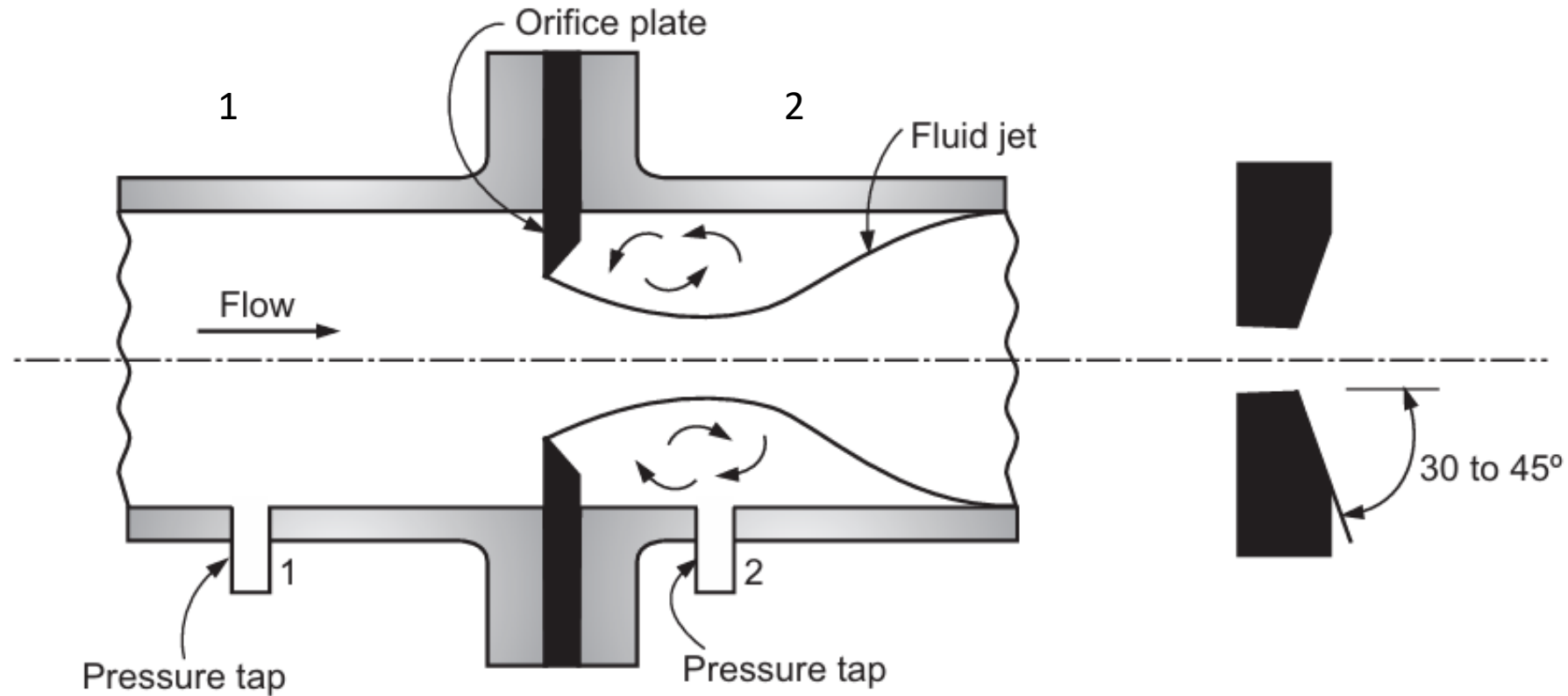


Fig. 7.25 : Orifice meter

<https://uorepc-nitk.vlabs.ac.in/exp/flow-through-orifice-meter/theory.html>

Orifice Meter

Construction :

- An orifice meter consists of a flat circular plate with an accurately machined and drilled circular hole called an orifice.
- The orifice diameter may vary from 0.2 to 0.85 times the pipe diameter, but generally the orifice diameter of 0.5 times the diameter of pipe is used.
- Pressure taps are provided, one each on the upstream side (station 1) and downstream side (station 2) of the orifice plate where a converging jet of the fluid has a minimum cross-sectional area.
- This point of minimum cross-section of flow is known as the **vena contracta** resulting in almost a maximum velocity and consequently a minimum pressure at this section.
- This pressure difference is measured by connecting a manometer or other pressure measuring device, e.g., pressure gauges to the pressure taps.

Orifice Meter

Working:

- In the meter, the fluid is first accelerated and then retarded so that a pressure drop across the meter is created.
- Once the steady state is attained, the pressure drop across the meter is noted (for a U-tube manometer, it is a manometer reading and for pressure gauges, it is a difference in the reading of the pressure gauges on both sides of the meter).
- This pressure drop is then used to calculate the volumetric flow rate using a mathematical flow equation for the meter.

Orifice Meter

Applying Bernoulli's equation
between point 1 and 2

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$Z_1 = Z_2$$
$$\Rightarrow \frac{P_1}{\rho g} + \frac{V_1^2}{2g} = \frac{P_2}{\rho g} + \frac{V_2^2}{2g}$$

$$\frac{P_1}{\rho g} - \frac{P_2}{\rho g} = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$
$$\frac{P_1 - P_2}{\rho g} = \frac{V_2^2 - V_1^2}{2g}$$

$$\Delta P = \rho g h$$
$$h = \frac{\Delta P}{\rho g} = \frac{P_1 - P_2}{\rho g}$$

$$h = \frac{V_2^2 - V_1^2}{2g} \quad \text{--- (1)}$$

Orifice Meter

A_o = area of orifice
 C_c = coefficient of contraction

$$C_c = \frac{A_2}{A_o} \rightarrow A_2 = C_c A_o$$

Applying continuity equation
 $A_1 V_1 = A_2 V_2$

$$V_1 = \frac{A_2 V_2}{A_1}$$

$$V_1 = \frac{C_c A_o V_2}{A_1}$$

Orifice Meter

substituting v_1 in (1)

$$h = \frac{v_2^2 - \left(\frac{C_c A_0 v_2}{A_1} \right)^2}{2g}$$
$$h = \frac{A_1^2 v_2^2 - C_c^2 A_0^2 v_2^2}{A_1^2 \times 2g}$$
$$2gh = \frac{(A_1^2 - C_c^2 A_0^2) v_2^2}{A_1^2 \times 2g}$$
$$v_2^2 = \frac{2gh}{\frac{A_1^2 - C_c^2 A_0^2}{A_1^2}}$$
$$v_2 = \sqrt{2gh} \sqrt{\frac{A_1^2}{A_1^2 - C_c^2 A_0^2}}$$
$$v_2 = A_1 \sqrt{\frac{2gh}{A_1^2 - C_c^2 A_0^2}}$$

Orifice Meter

$$Q = A_2 v_2$$

$$Q_{th} = A_1 A_2 \sqrt{\frac{2gh}{A_1^2 - C_c^2 A_0^2}}$$

$$Q_{ac} < Q_{th}$$

$$Q_{ac} = C_d \times Q_{th}$$

C_d = coefficient of discharge for orifice meter

$$Q_{ac} = C_d A_1 A_2 \sqrt{\frac{2gh}{A_1^2 - C_c^2 A_0^2}}$$

Orifice Meter

- The **value of the coefficient of orifice meter C_d is about 0.61** (for Reynolds numbers more than 10^4) indicating that there is a substantial loss in pressure over the instrument (i.e., the pressure recovery in this meter is very poor) which is the most serious disadvantage of the meter.
- The flow lines run parallel and widely spaced on the upstream side of the plate but as the flow approaches the orifice plate, the flow lines slowly converge at the orifice and then diverge out to pipe area smoothly.
- Since the flow lines converge and then diverge, the flow area becomes minimum at a short distance on the downstream side of the orifice plate which is known as the vena-contracta.
- The jet of fluid issuing from the orifice plate gradually expands from the vena-contracta to again fill the entire cross-section of the pipe.
- In this meter, since there is a sudden change in the cross-sectional area of the flow passage and not a gradual change in the cross-sectional area of the flow passage as that in the venturi meter, there is a greater loss of energy in the orifice meter than in the venturi meter.

Orifice Meter

- Orifice meters are used for testing purposes and in steam lines, where the power loss is of less importance.

Advantages of Orifice meter :

1. Simple in construction. It is easy to install.
2. Installing a new orifice plate with different opening is a simple job.
3. Greater flexibility.
4. It can be used with all types of differential pressure transmitters.
5. It is cheap.
6. The ratio of orifice diameter to pipe diameter can be changed.

Disadvantages of Orifice meter :

1. Substantial loss of pressure over the instrument and thus the pressure recovery is poor.
2. Considerable power loss.
3. It cannot be used for slurries that may clog the orifice opening.
4. For accurate readings, it requires a certain straight run of pipe without any pipe fittings or a valve on both sides of the meter.

DIFFERENCE BETWEEN ORIFICE METER AND VENTURI METER

Orifice meter	Venturi meter
1. Simple in construction.	1. Relatively complex in construction.
2. Low space requirement.	2. Occupies considerable space.
3. Relatively cheap.	3. Expensive.
4. Ratio of orifice diameter to pipe diameter can be changed simply by installing a new orifice plate.	4. Ratio of throat diameter to pipe diameter cannot be changed.
5. Pressure recovery is very poor.	5. Pressure recovery is very high.
6. Coefficient of discharge is about 0.61.	6. Coefficient of discharge is about 0.98.
7. Larger power loss.	7. Smaller power loss.
8. Fluid is retarded at once.	8. Fluid is retarded gradually.
9. Flow separates from the wall, i.e., flow separation is there.	9. No flow separation (possibility very less).
10. Area is minimum at the vena contracta.	10. Area is minimum at the throat and there is no vena contracta.
11. Coefficient of contraction is not unity.	11. Coefficient of contraction is unity.
12. Cannot be used where only small pressure head is available.	12. Can be used where only a small pressure head is available.

- <https://www.youtube.com/watch?v=bafyNbgx-sE>
- <https://www.youtube.com/watch?v=5hiwku3Z5m4>

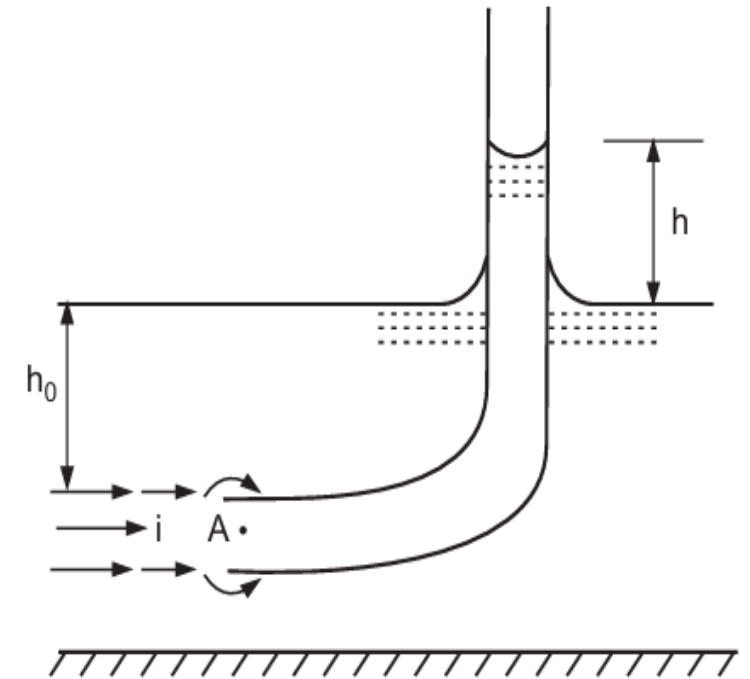
Pitot Tube

- It is a device used to measure the local or point velocity by measuring the difference between impact pressure and static pressure.
- The orifice meter and venturi meter measure the average velocity of the entire fluid stream, whereas the pitot tube measures the velocity at one point only.
- **Principle:**
- The basic principle is that if the velocity of flow at a particular point is reduced to zero (known as the stagnation point), the pressure at that point increases due to conversion of the kinetic energy into the pressure energy, and by measuring the increase in the pressure energy at this point the velocity may be determined.

Pitot Tube

Construction and Working :

- It consists of a glass tube, sufficient large for capillary effects to be negligible and bent at right angles.
- A single tube of this kind may be used for measuring the flow velocity in an open channel.
- The tube is dipped vertically in the flowing stream of fluid with its open end A facing the flow and the other open end projecting above the liquid surface.
- The fluid enters the tube and the level of the fluid in the tube exceeds that of the fluid surface.
- At the stagnation point, the kinetic energy is converted into the pressure energy and hence the fluid in the tube rises above the surrounding fluid surface by a height that corresponds to the velocity of fluid approaching the end A of the tube.



Pitot Tube

Consider a point 1 at the upstream of end A lying in a horizontal plane in the flowing stream where the flow velocity is u .

Applying the Bernoulli's equation between points 1 and A, we get

$$h_o + u^2/2g = h_o + h$$

where h_o is the height of the fluid raised in the tube above the free surface

$(h_o + h)$ is the stagnation pressure head at point A which consists of a static pressure head h_o and dynamic pressure head/impact pressure head h .

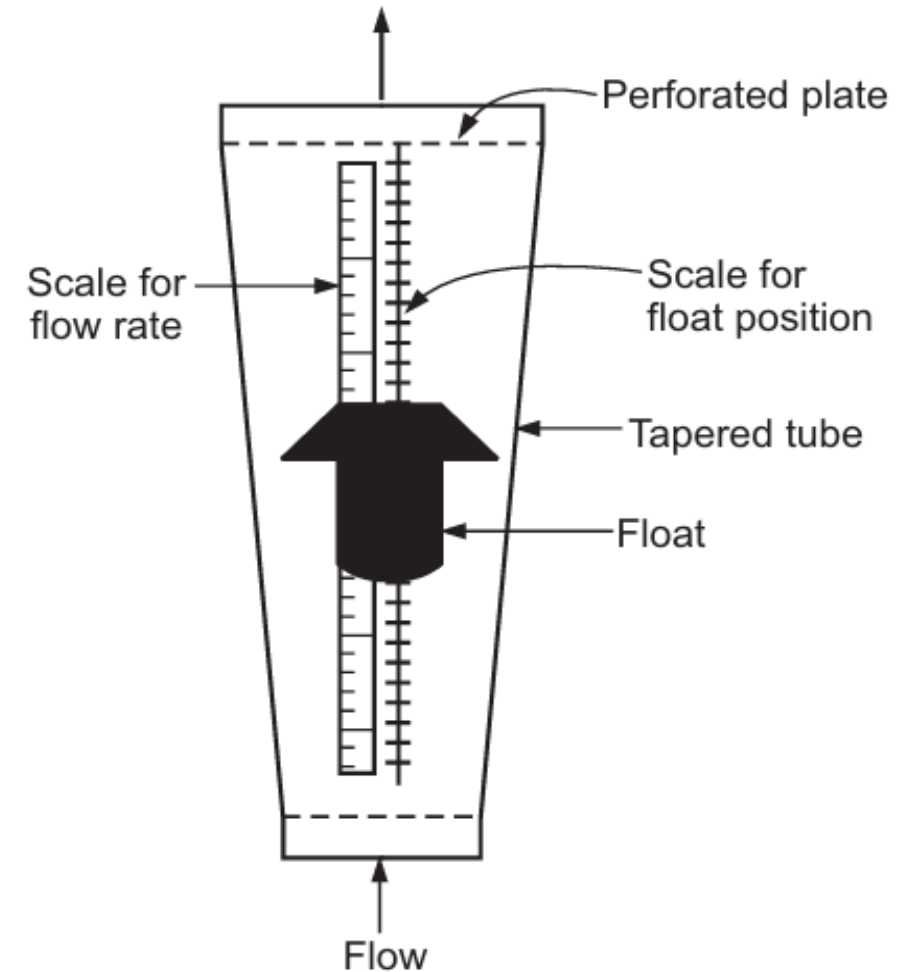
$$u = \sqrt{2gh}$$

Taking into account the loss of energy, Equation is modified to yield the actual flow velocity, where **C is the coefficient of pitot tube and it is generally taken as 0.98.**

$$u = C \sqrt{2gh}$$

Rotameter (Variable Area Meter)

- **Principle :** Rotameter operates on the principle that there is a different constriction / orifice area (flow area) for each flow rate and the pressure drop across the meter is constant.
- In other words, the pressure drop across the meter is constant and the area through which the fluid flows varies with the flow rate.
- This area is related, through a proper calibration, to the flow rate.



Rotameter (Variable Area Meter)

- **Construction :**
- It consists of a tapered glass tube mounted vertically in a frame with the large end up. The tube contains a freely moving solid float.
- The diameter of the float is smaller than the diameter of the bottom of the tapered tube.
- The density of a float material is higher than that of the liquid.
- The float is usually constructed out of corrosion resistance materials such as stainless steel (SS – 304 or SS – 316), aluminium, monel, bronze, nickel and plastics.
- A nearly linear flow scale is etched/marked on the glass tube or it is mounted close to the tube so that the position of the float can be marked and the flow rate is then obtained from a calibration curve or a direct scale of the flow rate can also be provided over the tapered tube or near the tube.
- The float is the indicating element and the reading edge of the float is taken at the largest cross-section of the float.

Rotameter (Variable Area Meter)

- The either ends of the meter may be screwed or flanged.
- Rotameters do not require straight runs of pipe before and after the point of installation.
- For low pressure and temperature applications, glass tubes are used but for opaque liquids, high temperature and pressure applications or for other conditions where glass is impracticable (not suitable), metal tubes are used.
- When a metal tube is used (the float in this case is invisible) or when a liquid is very dark or dirty, an external indicator is required (for the meter reading).

Rotameter (Variable Area Meter)

Working :

- In rotameter as the flow varies, the float rises or falls, thus changing the area of the annular space between the float and the wall of the tube.
- When no fluid flows through the meter, the rotameter float rests at the bottom of the tube.
- But as the fluid begins to flow from the lower side of the tube, the float rises until the differential pressure just balance the weight of the float and the fluid flows through the meter through the annular space.
- As the flow rate increases, the float rises further in the tube, thus, increasing the area available for flow (flow area) keeping a differential pressure across it constant.
- At a given flow rate, float stabilises at a certain fixed position in the tube and at steady state it is recorded as the rotameter reading on the scale provided.
- The variation of the flow area with flow rate can be measured in terms of change in the float position.
- Rotameters are widely used in the chemical industry for the measurement of flow rates of compressible as well as incompressible fluids.

DIFFERENCE BETWEEN VARIABLE HEAD METERS AND VARIABLE AREA METERS

Variable Head Meter	Variable Area Meter
1. In variable head meters, the area of constriction/area of flow is constant and the pressure drop varies with the flow rate.	1. In variable area meters, the pressure drop is constant and the area of flow/area of constriction varies with the flow rate.
2. Variable head meters include orifice meter and venturi meter.	2. Variable area meters include rotameters of various designs.
3. Simple in construction.	3. Complex in construction.
4. Relatively cheaper.	4. Relatively costly.
5. Needs straight runs of pipe before and after the meter (i.e., without fittings, valve).	5. Does not need straight runs of pipe before and after the meter.
6. It cannot give direct visual reading of the flow rate.	6. It can give direct visual reading of the flow rate.

Measurement of Flow in Open Channels

- A notch may be defined as an opening provided in the side of a tank/vessel/reservoir such that the liquid surface in the tank (i.e., the upstream liquid level) is below the top edge of the opening.
- A notch may have only the bottom edge and sides because its top edge above the liquid surface/level serves no purpose.
- The bottom edge of a notch, over which the liquid flows, is known as sill or crest of the notch.
- Usually, a notch is made of a metallic plate. It is used to measure the rate of discharge of liquids/the rate of flow of liquids flowing in open channels, not in closed pipelines.

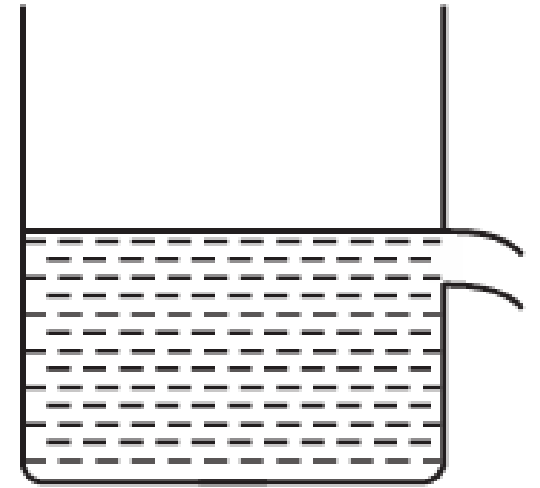


Fig. 7.38 : A notch

Measurement of Flow in Open Channels

- The notches are usually classified according to their shapes :

i. Rectangular notch

$$Q = \frac{2}{3} C_d B \sqrt{2g} (H)^{3/2}$$

B is the width or length of the notch and C_d is the coefficient of discharge (0.60), H is the height of liquid sill/crest of the notch

ii. Triangular notch/V-notch

θ be the angle of the notch

$$Q = \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{5/2}$$

For a 90° notch for which $C_d = 0.6$ and for $g = 9.81 \text{ m/s}^2$,

$$Q = 1.417 H^{5/2} = 1.417 (H)^{2.5} .$$

iii. Trapezoidal notch

Unit 5: Transportation and Metering

Transportation of fluids:

- Fluid moving machinery performance
- Selection and specification
- Positive displacement pumps
- Rotary and Reciprocating pumps
- Centrifugal pumps and characteristics
- Introduction to Fans, Blowers & Compressors

Transportation of Fluids

- In the chemical process industries for the transportation of fluids from one process equipment to another, or through long pipes or ducts we have to use some form of a mechanical pumping device.
- Such a mechanical device may be a pump, fan, blower or compressor, which increases the fluid energy.
- A pump is a machine which converts mechanical energy supplied to it from some external source into pressure energy which is used to lift a liquid from a lower level to a higher level.
- For the transportation (handling) of liquids, pumps are used, whereas for the transportation of gases, fans, blowers or compressors are used.

Factors which influence the choice of pump

1. The quantity of liquid to be handled.
2. The head against which the liquid to be pumped/raised.
3. The nature of liquid to be handled (viscosity, clear liquids, suspensions, corrosive nature).
4. The nature of power supply.
5. The method of operation – continuous or intermittent.
6. The flow rate required.
7. The pressure on delivery.
8. Cost and mechanical efficiency.

Measurement of performance

Capacity :

- It is the rate at which the fluid is pumped by a fluid moving device.
- It is expressed in various units – based on the type of pumping device.
- For liquids, it is expressed in litres per minute or gallons per minute and for gases, it is expressed in cubic feet per minute at inlet conditions of gases to the machine.

Overall efficiency :

- It is the ratio of the useful hydraulic work performed to the actual work input, irrespective of the type of drive.

Transportation of Fluids

CLASSIFICATION OF PUMPS:

1. Centrifugal pumps:

- The kinetic energy of the liquid to be pumped is increased by the action of centrifugal force and is then converted into pressure energy that is needed to pump the liquid.

2. Positive displacement pumps:

- A definite or fixed quantity of liquid is pumped in every revolution. Since a fixed quantity of liquid is pumped after each revolution they can develop a very high pressure if the delivery valve or line is closed.
- In reciprocating pumps, the displacement is by reciprocation of a piston or plunger, while in rotary pumps, the displacement is by rotary action of a gear or vane.
- Positive displacement pumps are further classified as : 1. Reciprocating pumps, e.g., piston pump, plunger pump, etc. 2. Rotary pumps, e.g., gear pump, mono pump.

Centrifugal Pumps

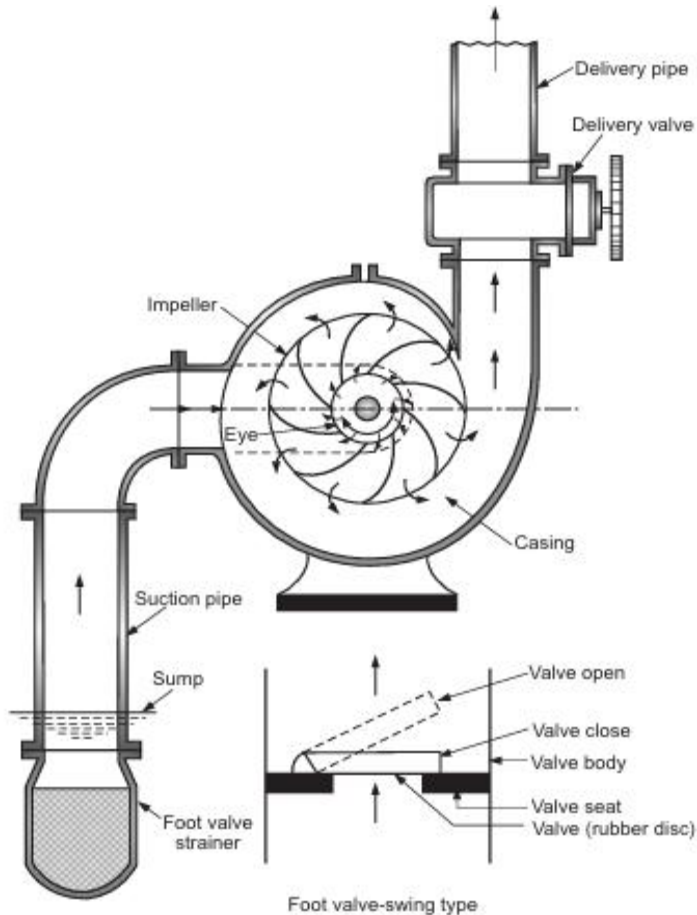


Fig. 8.3 : Component parts of a centrifugal pump

Working:

1. A pump which lifts a liquid from a lower level to a higher level by the action of a centrifugal force is called as a centrifugal pump.
2. In a centrifugal pump, power from an outside source is applied to the shaft, the impeller then rotates within the stationary casing.
3. The rotation of impeller blades produces a decrease in pressure at the eye of the impeller.
4. Due to this, liquid flows into the impeller from the suction pipe.
5. The liquid is then thrown outward by the centrifugal action along the blades.
6. As a result of high speed of rotation, the liquid acquires a high kinetic energy.
7. The kinetic energy acquired is then converted into pressure energy when it leaves the blade tips and the liquid passes into the volute chamber and finally it is discharged through the outlet (discharge) on the pump.

Centrifugal Pumps

- In the operation of a centrifugal pump before the pump is started, priming of the pump is done.
- In the priming operation, the suction pipe, pump casing, and portion of the delivery pipe upto a delivery valve are completely filled with the liquid to be pumped so that all the air, gas or vapour from this portion of the pump is expelled out and no air pocket is left.
- In presence of even very small air pocket in any of these portions, pump will not discharge the liquid.
- The need to do priming of pump is due to the fact that the pressure generated by a centrifugal pump impeller is directly proportional to the density of fluid that is in contact with it.
- Therefore, if the impeller is rotated in the presence of air, only negligible pressure would be produced and thus no liquid will be lifted by the pump.
- After the pump is primed properly, the delivery valve is kept close and power from an outside source (electric motor) is applied to the shaft.
- The impeller then rotates within the stationary casing. The rotation of the impeller produces a forced vortex which imparts a centrifugal head to the liquid and thus results in an increase of pressure throughout the liquid mass.

Advantages of Centrifugal Pump :

1. It is simple in construction.
2. Due to its simplicity of construction, it can be made in a wide range of materials.
3. Low initial cost and simplicity of design.
4. It operates at high speed and hence, can be coupled directly to an electric motor. In general, higher the speed, smaller the pump and motor required for a given duty.
5. It gives a steady delivery / discharge.
6. Lower maintenance (compared to other pumps) costs.
7. It does not get damaged even if the delivery line becomes blocked, or the delivery valve is closed, provided the pump does not run in this condition for a prolonged period.
8. It can handle readily liquids containing high proportions of suspended solids.
9. For equal capacity, the centrifugal pump is much smaller than any other type of pump. Therefore, it can be made into a sealed unit with the driving motor and immersed in the suction tank.

Disadvantages of Centrifugal Pump :

1. It is not usually self-priming.
2. It operates at low efficiencies (50 – 65%).
3. It cannot handle very viscous liquids efficiently.
4. It does not develop a high pressure.
5. Multistage pumps will develop greater pressure heads but they are much more expensive and cannot be made into corrosion resistant materials because of their greater complex construction.
6. If a non-return valve is not provided in the delivery or suction line, the liquid will run back into the suction tank (reservoir) as soon as the pump stops.

Head of a Centrifugal Pump

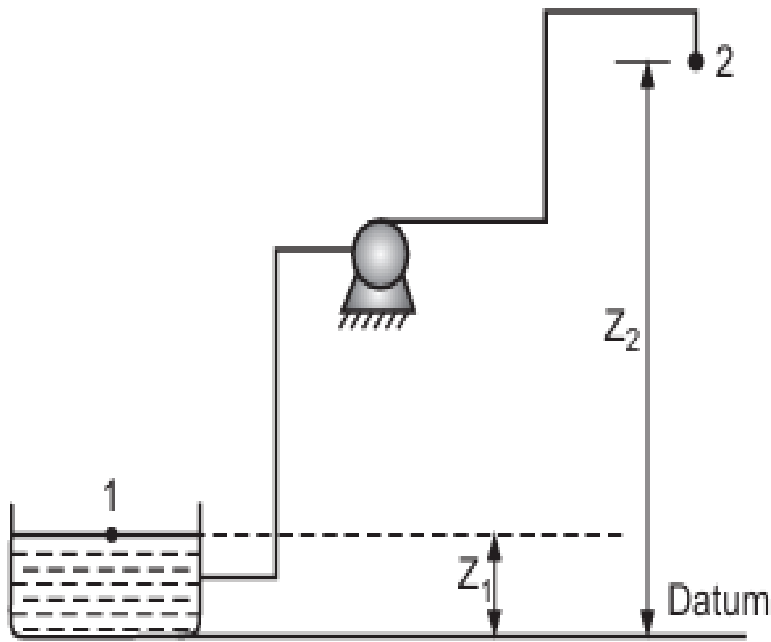


Fig. 8.7 : Pump flow system

- Total head developed by a pump is composed of the difference between static pressures and velocity heads plus the frictional losses in the suction and discharge line of the pump.
- Let Z_1 , Z_2 be the elevations of stations 1 and 2 respectively above a certain datum level. u_1 and u_2 be the velocities and P_1 and P_2 be the pressures at stations 1 and 2 respectively.
- Let W_p be the pump work per unit mass of fluid and η be the (mechanical) efficiency of the pump.

$$\begin{aligned} \eta W_p / g &= \left(Z_2 + \frac{u_2^2}{2g} + \frac{P_2}{\rho g} + h_f / g \right) - \left(Z_1 + \frac{u_1^2}{2g} + \frac{P_1}{\rho g} \right) \\ &= H_2 - H_1 = \Delta H = \text{total developed head} \end{aligned}$$

Cavitation

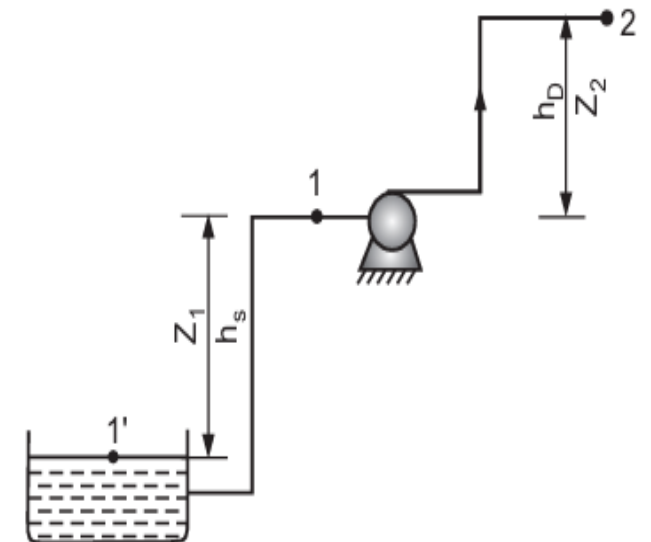
- If the pressure in the suction line is less than the vapour pressure of the liquid, then some of the liquid flashes into vapour. This phenomenon is called cavitation.
- When cavitation occurs the liquid will not be pumped.
- Therefore, care must be taken so that the pressure in the suction line does not fall below the vapour pressure of the liquid to be pumped.
- Cavitation will not take place when the sum of the velocity and suction heads at the suction is sufficiently greater than the vapour pressure of the liquid at the temperature of pumping.
- The cavitation can be eliminated by avoiding sharp bends in the suction line to reduce loss of head, keeping velocity in the suction line low and using low pumping temperatures to keep the vapour pressure low.
- The occurrence of cavitation can be noted by a marked increase in noise and vibration and the cavitation leads to the mechanical damage to the pump as the vapour bubbles collapse.

Net Positive Suction Head (NPSH)

- The NPSH is the amount by which the pressure at the suction point of the pump, expressed as a head of the liquid to be pumped (the sum of the velocity and pressure heads), is in excess of the vapour pressure of the liquid.
- If the required value of NPSH is not obtained, then partial vaporisation is likely to take place and both suction and delivery heads get reduced.

$$\text{NPSH} = \left(\begin{array}{c} \text{absolute pressure head} \\ \text{available at suction point} - 1 \end{array} \right) - (\text{vapour pressure head})$$
$$= \frac{u_1^2}{2g} + \frac{P_1}{\rho g} - \frac{P_v}{\rho g}$$

- Z_1 is the static suction lift (denoted by h_s)
- Z_2 is the static delivery lift (denoted by h_D)
- P_v = vapour pressure of the liquid at the pumping temperature



Net Positive Suction Head (NPSH)

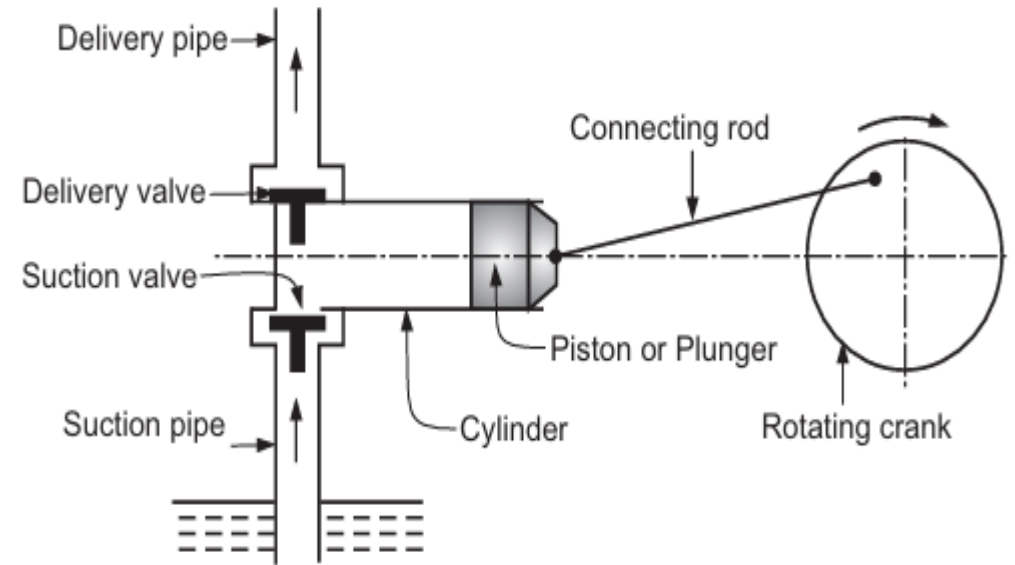
- If h_{fs} is the head lost in friction in the suction line. If $Z'_1 = 0$ and $u'_1 = 0$, then

$$\text{NPSH} = \frac{P'_1}{\rho g} - \frac{P_v}{\rho g} - Z'_1 - h_{fs}$$

- If $\text{NPSH} = 0$, the cavitation will occur as at this value the suction pressure equals the vapour pressure of the liquid.
- NPSH must be greater than zero. The required value of NPSH is usually 2 or 3 m or more.
- NPSH may also be defined as the head required to make the liquid flow through the suction pipe to the impeller.

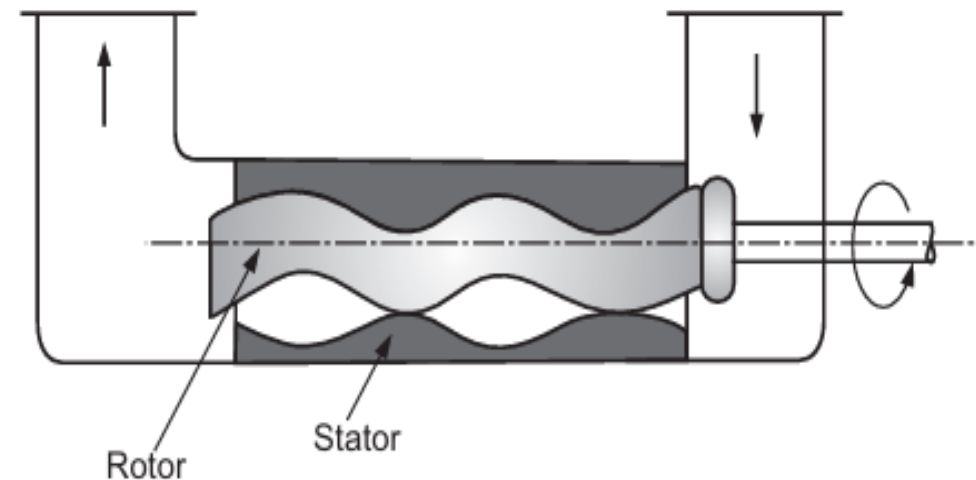
Positive Displacement Pumps - Reciprocating Pumps

- Reciprocating pump essentially consists of a piston or plunger or diaphragm which moves to and fro i.e., reciprocates back and forth in a close fitting stationary cylinder.
- The cylinder is connected to suction and delivery pipes.
- The suction valve allows the liquids only to enter the cylinder and the delivery valve allows only its discharge from the cylinder.
- A piston or plunger is connected to a crank by means of a connecting rod. The crank is rotated by a driving engine or electric motor.
- When the crank is rotated by the drive, the piston or plunger moves to and fro (forward and backward movement) in the cylinder.
- The reciprocating pumps are provided with an air vessel at the discharge.



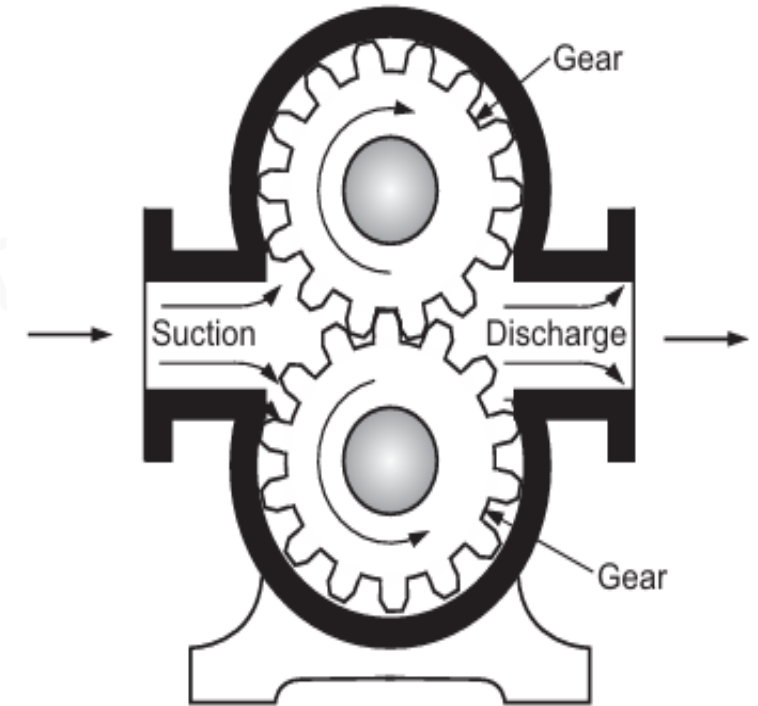
Positive Displacement Rotary Pump - Mono Pump

- This pump consists of a rotor that rotates in a stator made of rubber or other similar material. The rotor is a true helical metal screw while the stator has a double helical thread pitched opposite to the spiral on the rotor. The liquid to be pumped moves continuously towards a discharge through the voids between the rotor and the stator.
- This pump is capable of handling highly viscous materials, gritty liquids, etc.
- It is widely used in the chemical industry for feeding slurries containing higher proportion of solids to filtration equipments.



Positive Displacement Rotary Pump - Gear Pump

- It consists of two toothed gear wheels enclosed in a casing provided with inlet and outlet connections for liquid to be pumped.
- Of the two gear wheels, one is driven by an electric drive and other rotates in mesh with it.
- The liquid entering in the pump from the inlet due to creation of a reduced pressure at the inlet is carried round in the space between the gear teeth and the casing during the rotation of gear wheels and after further rotation the liquid is pumped out of the discharge side as the teeth come into mesh.
- The gear pumps are commonly used in the chemical industry for handling high viscosity liquids including molasses, paints and greases.
- The gear pump is not suited for liquids that carry solids in suspension due to close clearance between gear wheels and teeth.



Centrifugal Pump	Reciprocating Pump
1. The means of fluid transfer is a centrifugal force. 2. Simple in construction and light in weight due to less number of parts. 3. It requires less floor space and simple foundation. 4. It needs priming. 5. It does not incorporate an air vessel. 6. Discharge is steady and even. 7. Used for large capacity and low heads. 8. It is operated at high speed. 9. It can handle relatively viscous liquids and liquids containing suspended solids. 10. They are coupled directly to an electric motor. 11. They are operated against a closed valve without danger. 12. Efficiencies are low. 13. Maintenance cost is low. 14. Costwise they are cheaper. 15. Designed for high discharge. 16. Cannot develop high pressures.	1. The means of fluid transfer is a displacement of volume by a piston element. 2. Complex in construction and heavy construction due to more number of parts. 3. It requires more floor space and comparatively heavy foundation. 4. It does not need priming. 5. It incorporates an air vessel. 6. Discharge is not even, it is pulsating. 7. Used for low capacity and high heads. 8. It is operated at low speed. 9. It cannot handle viscous liquids and liquids containing suspended solids. It can handle clear liquids. 10. They are belt driven. 11. They are never operated against a closed valve. 12. Efficiencies are high. 13. Maintenance cost is high. 14. Cost of the reciprocating pump is higher than the centrifugal pump of the same power. 15. Designed for high heads. 16. Can develop high pressures.

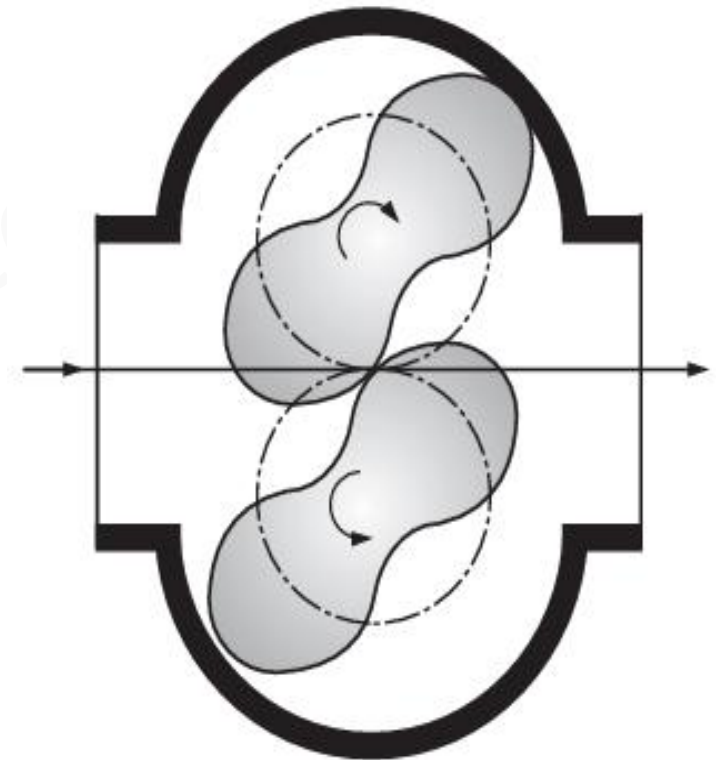
Pumping Devices for Gases - Fans, blowers, and compressors

FANS:

- Fans are used for moving gases when the pressure heads of less than 30 kPa are involved.
- They are either centrifugal type or axial flow type.
- The centrifugal type fans depend on the centrifugal force for propelling the gas, while axial flow type fans impart energy to the gas as it flows parallel to the central axis of the fan.
- The inlet and outlet volumes of gases are essentially equal due to very low operating pressures and hence are simply movers of gas.
- Fans are employed industrially for ventilation works, supplying air to dryers, supplying draft to boilers, removal of fumes, etc.

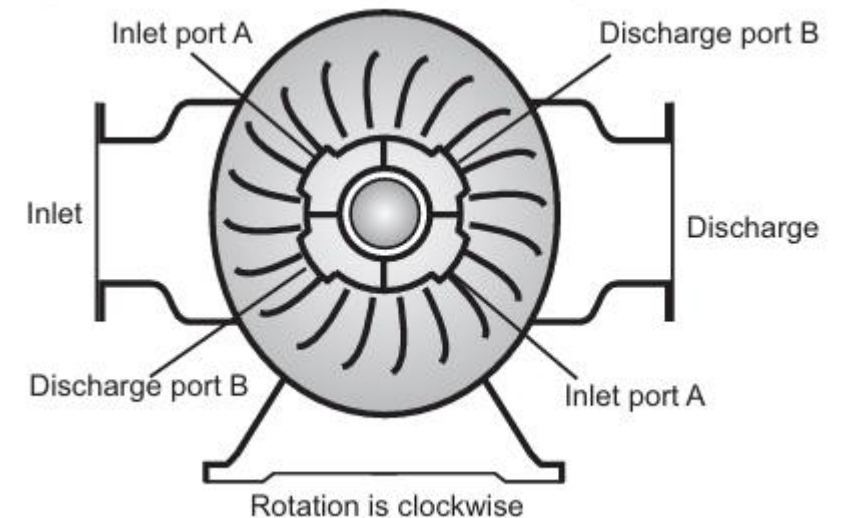
Cycloidal / Lobe / Rootes Blower

- It is a positive displacement blower, working similar to gear pump.
- This type of blower is used for capacity upto $7 \text{ m}^3/\text{s}$ and discharge pressure upto 90 kPa.
- It is usually used for services where very large volumes must be delivered against a pressure too high for fans.
- It consists of two lobe rotors incorporated in a casing. The casing is provided with inlet and outlet connections for the gas to be handled.
- The compression is achieved by the rotation of two rotors. The rotors turn in opposite directions, each passes to the inlet, takes in the gas, it is compressed between impeller and the wall (casing) and finally it is discharged outward.



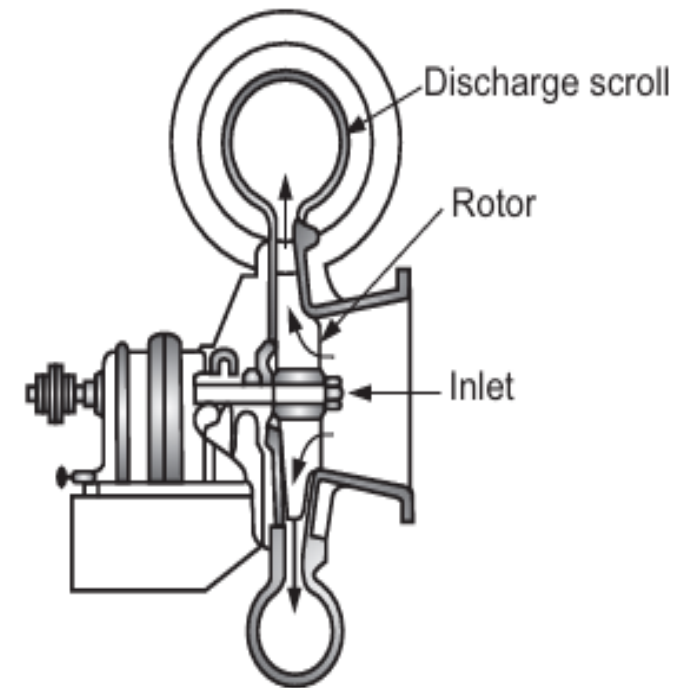
Nash Hystor Pump

- It is a positive displacement blower (rotary) and is also known as liquid piston type of rotary blower.
- It consists of a vaned impeller incorporated centrally in a casing. The casing is approximately elliptical in shape.
- The inlet and discharge ports are located in the impeller hub.
- A sealing liquid is filled inside the casing to seal the impeller at its points of least clearance from the casing.
- As the impeller rotates, the centrifugal force drives the sealing liquid against the walls of the elliptical casing.
- The sealing liquid alternately moves towards and go back (away) from the centre of the impeller, thus acting essentially as a series of liquid pistons.
- The gas is sucked in from the inlet ports A as these recede from impeller and the gas is compressed into the outlet ports B as they advance towards the impeller.
- These units can be operated upto discharge pressure of 20 psi. These units are offered as single-stage units for pressure differential upto about 75 psi in the smaller sizes.



Centrifugal Blowers

- The principle of operation of a centrifugal blower is the same as that of the centrifugal pump.
- It consists of a vaned impeller keyed to a shaft connected to a drive.
- Units necessitate high speed of operation and large impeller diameter as very high heads in terms of low density fluid are needed to generate moderate pressure ratios.
- When the shaft rotates, the impeller blades are driven inside the casing. The impeller is surrounded by a diffuser.
- The function of the diffuser is to convert the kinetic energy of the gas leaving the impeller into pressure energy.
- The section of each impeller and its diffuser forms a stage.
- These units can be made to develop pressures of 275 kPa to 700 kPa with a multistage construction. The multistage construction is employed when high discharge pressure is required.
- Most centrifugal blowers/turbo blowers operate at speed 3500 rpm - 30,000 rpm.
- Centrifugal / turbo blowers are used for supplying air to furnaces, cooling and drying purposes, for transporting materials, for ventilation purposes.



Centrifugal Compressors

- Centrifugal compressors are multistage units containing a series of impellers on a single shaft which rotates at high speed in a massive (large and heavy) casing.
- In this machine, a gas flows into the eye of impeller, where it is accelerated radially, leaving at high velocity at the outer edge, and flowing into a diffuser which converts kinetic energy into the pressure energy.
- The gas is then directed into the eye of the next impeller.
- Centrifugal compressors compress very large volumes of air or process gas upto $5400 \text{ m}^3/\text{min}$ at inlet to an outlet pressure of 2 MPa.
- The smaller capacity machines can produce discharge pressures upto several hundred atmospheres.
- These machines usually include some provision for cooling.
- Centrifugal compressors are widely used in petroleum refineries and chemical plants.

Reciprocating Compressors

- Reciprocating compressors are available either as a single-stage or multistage units for pressures as high as 240 MPa.
- A reciprocating compressor incorporates a piston, a cylinder with intake and exhaust valves and a crank shaft with a drive.
- These units operate mechanically in the same manner as the reciprocating pumps and are usually double acting.
- The cylinder of the compressor is usually water jacketed to remove heat of compression. The reciprocating compressors are usually belt driven from an electric motor.
- When it is not possible to achieve the required compression ratio with single stage units, multistage units are used. In case of multistage compressors, it is general practice to cool the gas between the stages.

SQ Q4. A rectangular notch 2.5 m wide has a constant head of 0.40 m. Find the rate of discharge of water through the notch if the coefficient of discharge for the notch is 0.60.

Solution : We have,

$B = \text{Width of the notch} = 2.5 \text{ m}$
 $H = \text{Head of water} = 0.40 \text{ m}$
 $g = 9.8 \text{ m/s}^2$
 $C_d = \text{Coefficient of discharge} = 0.60$

The relation to be used is

$$\begin{aligned} Q &= \frac{2}{3} C_d B \sqrt{2g} (H)^{1.5} \\ &= \frac{2}{3} \times 0.60 \times 2.5 (2 \times 9.81)^{1/2} (0.40)^{1.5} \\ &= 1.120 \text{ m}^3/\text{s} \quad \text{or} \quad 1120 \text{ l/s} \end{aligned}$$

The rate of discharge of water is **1.12 m³/s or 1120 l/s**

SQ Q5. A rectangular notch with $C_d = 0.60$ has a discharge of 360 l/s when the head of water is half the length of the notch. Find the length of the notch.

Solution : Given : C_d = Coefficient of discharge = 0.60

Q = Rate of discharge = 360 l/s = 0.36 m³/s

B = Length/width of the notch

H = Head of water (depth) = Half the length of the notch = $B/2$

For the rectangular notch, Q is given by

$$Q = \frac{2}{3} C_d B \sqrt{2g} (H)^{1.5}, \text{ but } H = B/2$$

$$Q = \frac{2}{3} C_d B \sqrt{2g} (B/2)^{1.5} \quad B = 0.528 \text{ m}$$

$$0.36 = \frac{2}{3} \times 0.6 \times \sqrt{2 \times 9.81} \frac{B^{2.5}}{(2)^{1.5}} \quad \therefore (B)^{2.5} = 0.5747$$

$$\therefore B = 0.80 \text{ m} = 800 \text{ mm}$$

Length of the rectangular notch = **800 mm.**

... Ans.

SQ Q6. A pitot tube is a pipe in which air is flowing, is connected to a manometer containing water. If the difference in water levels in the manometer is 87.5 mm, what is the velocity of flow in the pipe assuming a tube coefficient of 0.99?

Solution : The velocity of flow is given by

$$u = c \sqrt{2g\Delta H}$$

where c = tube coefficient = 0.99

ΔH = manometer reading expressed in terms of m of the flowing fluid (air)

$\Delta H'$ = manometer reading in terms of water
 = 87.5 mm = 8.75 cm
 = 0.0875 m of H₂O

$$\Delta H = \Delta H' \left[\frac{\rho_{\text{H}_2\text{O}} - \rho_{\text{air}}}{\rho_{\text{air}}} \right]$$

where,

$$\rho_{\text{H}_2\text{O}} = 1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$$

$$\rho_{\text{air}} = 1.29 \times 10^{-3} \text{ g/cm}^3 = 1.29 \text{ kg/m}^3$$

$$\Delta H = 0.0875 \left[\frac{1000 - 1.29}{1.29} \right] = 67.74 \text{ m of air}$$

$$\begin{aligned} \text{Velocity of flow of air} = u &= 0.99 [2 \times 9.81 \times 67.74]^{1/2} \\ &= \mathbf{36.09 \text{ m/s}} \end{aligned}$$

... Ans.

In this problem, the density of air is taken as $1.29 \times 10^{-3} \text{ g/cm}^3$.

LQ Q7. Water flows at the rate of 0.147 m³/s through a 150 mm diameter orifice inserted in a 300 mm diameter pipe. If pressure gauges fitted at the upstream and the downstream of the orifice plate give readings of 176.58 kN/m² and 88.29 kN/m² respectively, find the coefficient of discharge of the orifice meter. Density of water = 1000 kg/m³.

Solution : The flow equation of an orifice meter is

$$Q = \frac{C_o A_o}{\sqrt{1 - \beta^4}} \times \frac{\sqrt{2 (P_a - P_b)}}{\rho}$$

$$D_o = 150 \text{ mm} = 0.15 \text{ m}, D = 300 \text{ mm} = 0.30 \text{ m}$$

$$\beta = \frac{D_o}{D} = \frac{150}{300} = 0.5$$

$$\text{Volumetric flow rate} = 0.147 \text{ m}^3/\text{s}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$P_a = 176.58 \text{ kN/m}^2 = 176580 \text{ N/m}^2$$

$$P_b = 88.29 \text{ kN/m}^2 = 88290 \text{ N/m}^2$$

$$A_o = \pi/4 D_o^2 = \pi/4 (0.15)^2 = 0.01767 \text{ m}^2$$

Rearranging the flow equation gives

$$\begin{aligned} C_o &= \frac{Q \sqrt{1 - \beta^4}}{A_o \times \sqrt{2 (P_a - P_b) / \rho}} \\ &= \frac{0.147 \sqrt{1 - (0.5)^4}}{0.01767 \sqrt{\frac{2 (176580 - 88290)}{1000}}} \\ &= 0.606 \\ &\simeq 0.61 \end{aligned}$$

Coefficient of discharge of the orifice meter = 0.61

LQ Q8. A venturimeter is to be fitted in a pipe of 250 mm diameter where pressure head is 7.6 m of flowing fluid and the flow rate is 8.1 m³/min. Determine the diameter of the throat. Take the coefficient of venturimeter as 0.96.

Solution : Q = Volumetric flow rate = 8.1 m³/min = 8.1/60 = 0.135 m³/s

D = 250 mm = 0.250 m, D_T = diameter of throat.

The flow equation for a venturi meter is

$$Q = \frac{C_v A_T \sqrt{2 g \Delta H}}{\sqrt{(1 - \beta^4)}}$$

where ΔH = 7.6 m , g = 9.81 m/s²

$$\beta = D_T/D = D_T/0.250 = 4 D_T$$

$$A_T = \text{area of throat} = \pi/4 D_T^2$$

$$\therefore 0.135 = \frac{0.96 \times \pi/4 \times D_T^2 \times \sqrt{2 \times 9.81 \times 7.6}}{\sqrt{1 - (4 D_T)^4}}$$

Solving, we get

$$\sqrt{1 - (4 D_T)^4} = 68.2 D_T^2$$

$$1 - (4 D_T)^4 = (68.2 D_T^2)^2$$

$$1 - 256 D_T^4 = 4651.24 D_T^4$$

$$\therefore D_T^4 = (1 / 4907.24)$$

$$D_T = (1 / 4907.24)^{0.25}$$

$$= 0.1195 \text{ m}$$

$$\text{Diameter of the throat} = D_T = 0.1195 \text{ m}$$

$$= \mathbf{119.5 \text{ mm}}$$

LQ Q9. Water flows through an orifice of 25 mm diameter installed in a 75 mm i.d. pipe line at the rate of 0.3 l/s. What will be the difference in level on a water manometer connected across the meter? The coefficient of discharge of the meter is 0.61.

Solution : Volumetric flow rate of water = $Q = 0.3 \text{ l/s} = 0.3 \times 10^{-3} \text{ m}^3/\text{s}$

$$D_o = 25 \text{ mm} = 0.025 \text{ m},$$

$$D = 75 \text{ mm} = 0.075 \text{ m}$$

$$\beta = D_o/D = 25/75 = 0.333$$

$$C_o = 0.61$$

$$\begin{aligned} A_o &= \text{area of orifice} = \pi/4 (0.025)^2 \\ &= 4.909 \times 10^{-4} \text{ m}^2 \end{aligned}$$

The flow equation of an orifice meter is

$$Q = \frac{C_o A_o}{\sqrt{1 - \beta^4}} \sqrt{2g \Delta H}$$

$$0.3 \times 10^{-3} = \frac{0.61 \times 4.909 \times 10^{-4}}{\sqrt{1 - (0.333)^4}} \sqrt{2 \times 9.81 \times \Delta H}$$

$$\therefore \Delta H = 0.0505 \text{ m of H}_2\text{O}$$

$$\therefore \text{Difference in the levels on the water manometer} = 0.0505 \text{ m} = \mathbf{50.5 \text{ mm...}}$$

LQ Q10. A right-angled V-notch was used to measure the discharge of a centrifugal pump. Find the discharge through the notch, if the depth of water at the notch is 200 mm. Take $C_d = 0.60$.

Solution :

$$\theta = \text{Angle of the notch} = 90^\circ$$

$$H = \text{Depth of water at the notch} = 200 \text{ mm} = 0.20 \text{ m}$$

$$C_d = \text{Coefficient of discharge} = 0.60$$

$$Q = \text{Discharge through the notch}$$

$$\begin{aligned} Q &= \frac{8}{15} C_d \sqrt{2g} \tan \frac{\theta}{2} (H)^{2.5} \\ &= \frac{8}{15} \times 0.60 \times \sqrt{2 \times 9.81} \tan \frac{90}{2} (0.20)^{2.5} \\ &= \mathbf{0.02535 \text{ m}^3/\text{s} \text{ or } 25.35 \text{ l/s}} \end{aligned}$$

Or : For a 90° V-notch for $C_d = 0.60$, we have

$$\begin{aligned} Q &= 1.417 H^{5/2} \dots \text{in SI units} \\ &= 1.417 (0.20)^{5/2} \\ &= \mathbf{0.02535 \text{ m}^3/\text{s} \text{ or } 25.35 \text{ l/s}} \end{aligned}$$

LQ Q11. Calculate the net positive suction head (NPSH) of a centrifugal pump using the following data : (i) Vapour pressure of the liquid = 26.66 kN/m² (ii) Distance between the level of liquid in the reservoir and suction line = 1.2 m. (iii) Density of the liquid = 865 kg/m³. (iv) Friction in the suction line = 3.5 J/kg. (v) Reservoir is open to atmosphere.

Solution : NPSH in terms of J/kg, i.e., in energy units is given by the following equation:

$$\text{NPSH} = \frac{P_a' - P_v}{\rho} - h_{fs} - g Z_a$$

where $g = 9.81 \text{ m/s}^2$

$\rho = \text{density of the liquid} = 865 \text{ kg/m}^3$

$h_{fs} = \text{friction in the suction line} = 3.6 \text{ J/kg}$

$Z_a = \text{distance between the level of liquid in the reservoir and the suction line} = 1.2 \text{ m}$

$P_v = \text{vapour pressure of the liquid} = 26.66 \text{ kN/m}^2$
 $= 26660 \text{ N/m}^2$

$P_a' = \text{pressure over the liquid surface}$
 $= 101325 \text{ N/m}^2$

Substituting the values of various parameters in the above equation gives

$$\begin{aligned} \text{NPSH} &= \left(\frac{101325 - 26660}{865} \right) - 3.5 - 9.81 \times 1.2 \\ &= 71.04 \text{ J/kg} \\ &= 71.04 / g = 71.04 / 9.81 = \mathbf{7.24 \text{ m}} \end{aligned}$$

... Ans.

LQ Q12. A liquid of density 1150 kg/m^3 is flowing from point A to point B which is 5 metres above the point A. The frictional losses in a pipeline of internal diameter 40 mm are 1 J/kg for a volumetric flow rate of $500 \text{ cm}^3/\text{s}$. If points A and B are at the atmospheric pressure and the velocity at the point A is zero, calculate the pump work done.

Solution :

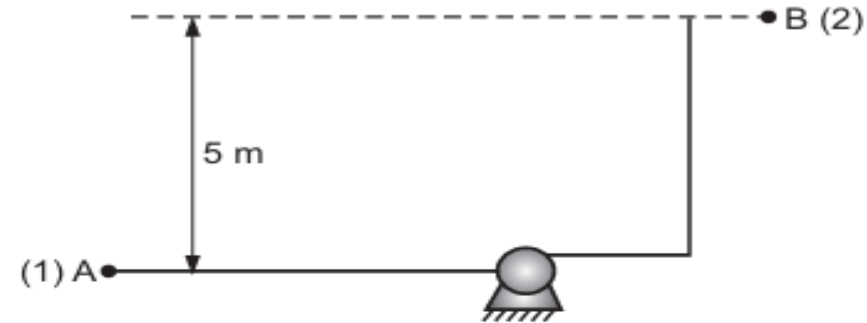


Fig. Ex. 8.4

Pressure at A = Pressure at B = 101.325 kPa

Density of the liquid = $\rho = 1150 \text{ kg/m}^3$

h_f = frictional losses = 1 J/kg

Volumetric flow rate = $Q = 500 \text{ cm}^3/\text{s}$

$$= 500 \times 10^{-6} \text{ m}^3/\text{s}$$

$$\text{Area of the pipe} = A = \frac{\pi}{4} D^2$$

where

$$D = 40 \text{ mm} = 0.04 \text{ m}$$

$$A = \frac{\pi}{4} (0.04)^2 = 1.26 \times 10^{-3} \text{ m}^2$$

Let us calculate u at B.

$$\begin{aligned}\text{Velocity at point B} &= u_2 = Q/A = \frac{500 \times 10^{-6}}{1.26 \times 10^{-3}} \\ &= 0.40 \text{ m/s}\end{aligned}$$

Given : Velocity at point A = $u_1 = 0$

Pressure at point A = P_1 ; Pressure at point B = P_2

The Bernoulli's equation between stations A and B is

$$\frac{P_1}{\rho} + g Z_1 + \frac{u_1^2}{2} + \eta W_p = \frac{P_2}{\rho} + g Z_2 + \frac{u_2^2}{2} + h_f$$

Let the point A (station 1) be at a datum line, then

$$Z_1 = 0, Z_2 = 5 \text{ m}, P_1 = P_2$$

With this the Bernoulli equation reduces to

$$\begin{aligned}\eta W_p = \text{pump work} &= u_2^2 / 2 + h_f + g Z_2 \\ &= (0.4)^2/2 + 1 + 9.81 \times 5 \\ &= \mathbf{50.13 \text{ J/kg}}\end{aligned}$$